

m3vcf.md

Here is what the list of files in the `m3vcf` subdirectory looks like. There are three files per non-sex chromosome:

- `ch*.phased.erate` : Error rate file which contains the per-marker phasing / genotyping error rates. This is auxiliary output for Minimac (reference panel pre-processing)
- `ch*.phased.m3vcf.gz` : These are the compressed reference haplotypes
- `ch*.phased.rec` : This is the recombination map used by an HMM during imputation

```
bash-4.4$ ls
chr1.phased.erate      chr12.phased.rec      chr16.phased.m3vcf.gz  chr2.phased.erate      chr22.phased.rec      chr6.phased.m3vcf.gz
chr1.phased.m3vcf.gz   chr13.phased.erate    chr16.phased.rec       chr2.phased.m3vcf.gz   chr3.phased.erate      chr6.phased.rec
chr1.phased.rec        chr13.phased.m3vcf.gz chr17.phased.erate     chr2.phased.rec        chr3.phased.m3vcf.gz   chr7.phased.erate
chr10.phased.erate     chr13.phased.rec       chr17.phased.m3vcf.gz  chr20.phased.erate     chr3.phased.rec        chr7.phased.m3vcf.gz
chr10.phased.m3vcf.gz  chr14.phased.erate    chr17.phased.rec       chr20.phased.m3vcf.gz  chr4.phased.erate      chr7.phased.rec
chr10.phased.rec       chr14.phased.m3vcf.gz  chr18.phased.erate     chr20.phased.rec       chr4.phased.m3vcf.gz   chr8.phased.erate
chr11.phased.erate     chr14.phased.rec       chr18.phased.m3vcf.gz  chr21.phased.erate     chr4.phased.rec        chr8.phased.m3vcf.gz
chr11.phased.m3vcf.gz  chr15.phased.erate    chr18.phased.rec       chr21.phased.m3vcf.gz  chr5.phased.erate      chr8.phased.rec
chr11.phased.rec       chr15.phased.m3vcf.gz  chr19.phased.erate     chr21.phased.rec       chr5.phased.m3vcf.gz   chr9.phased.erate
chr12.phased.erate     chr15.phased.rec       chr19.phased.m3vcf.gz  chr22.phased.erate     chr5.phased.rec        chr9.phased.m3vcf.gz
chr12.phased.m3vcf.gz  chr16.phased.erate    chr19.phased.rec       chr22.phased.m3vcf.gz  chr6.phased.erate      chr9.phased.rec
bash-4.4$ pwd
/project2/chia657_28/datasets/private/MEC/WGS/CCDG_WGS/pass_subset_10721_geno02_dropSingletons/m3vcf
```

What's in a `.m3vcf.gz`

The first 7 rows of the `.m3vcf.gz` look like:

```
##fileformat=M3VCF
##version=1.2
##compression=block
##n_blocks=7265
##n_haps=21442
##n_markers=882605
##<Note=This is NOT a VCF File and cannot be read by vcftools>
```

The 8th row contains: `#CHROM POS ID REF ALT QUAL FILTER INFO FORMAT` , followed by a list of tabbed haplotype ID's.

The haplotype ID's take the form of:

TWKG-EC000378_HAP_1
TWKG-EC000378_HAP_2
TWKG-EC000413_HAP_1
TWKG-EC000413_HAP_2
TWKG-EC000572_HAP_1
TWKG-EC000572_HAP_2
TWKG-EC000573_HAP_1
TWKG-EC000573_HAP_2
TWKG-EC000583_HAP_1
TWKG-EC000583_HAP_2

Where TWKG presumably refers to the Taiwan Genome Project reference panel, EC***** refers to some sort of anonymized participant ID, and HAP_1 or HAP_2 refer to the two haplotypes per individual.

There are 21442 unique haplotype IDs and 10721 unique participant IDs. This is true for all 22 autosomes.

TODO

- Remove individual (bug Bryan for information)
- Change EC IDs
- get Bryan's pipeline for m3vcf files
- Generate m3vcf files
 - Compare with previous results to see if there's the same